Week 2

Day 1

```
MON, 04/05/26
Ted Davis, P5LIVE –Offline,
```

START P5LIVE
cd /Users/chirin/Documents/P5LIVE
Npm start

STOP P5LIVE OFFLINE
CTRL + C

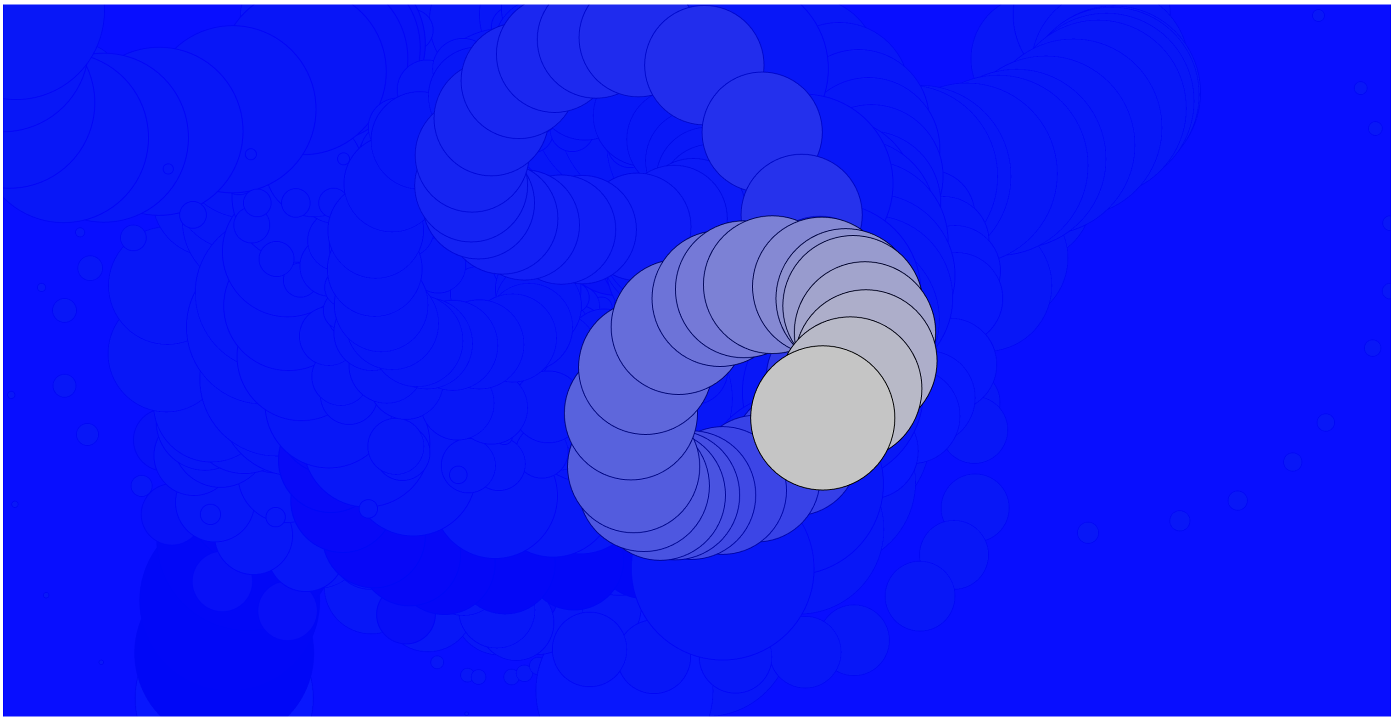
Ctrl + enter
Ctrl + shift + enter

```
function setup() {
  createCanvas(windowWidth, windowHeight)

}

function draw() {
  background(0,15,255,15) // r,g,b,a
  fill(frameCount % 255)
  circle(mouseX,mouseY,frameCount % 200)

  print(frameCount)
}
```

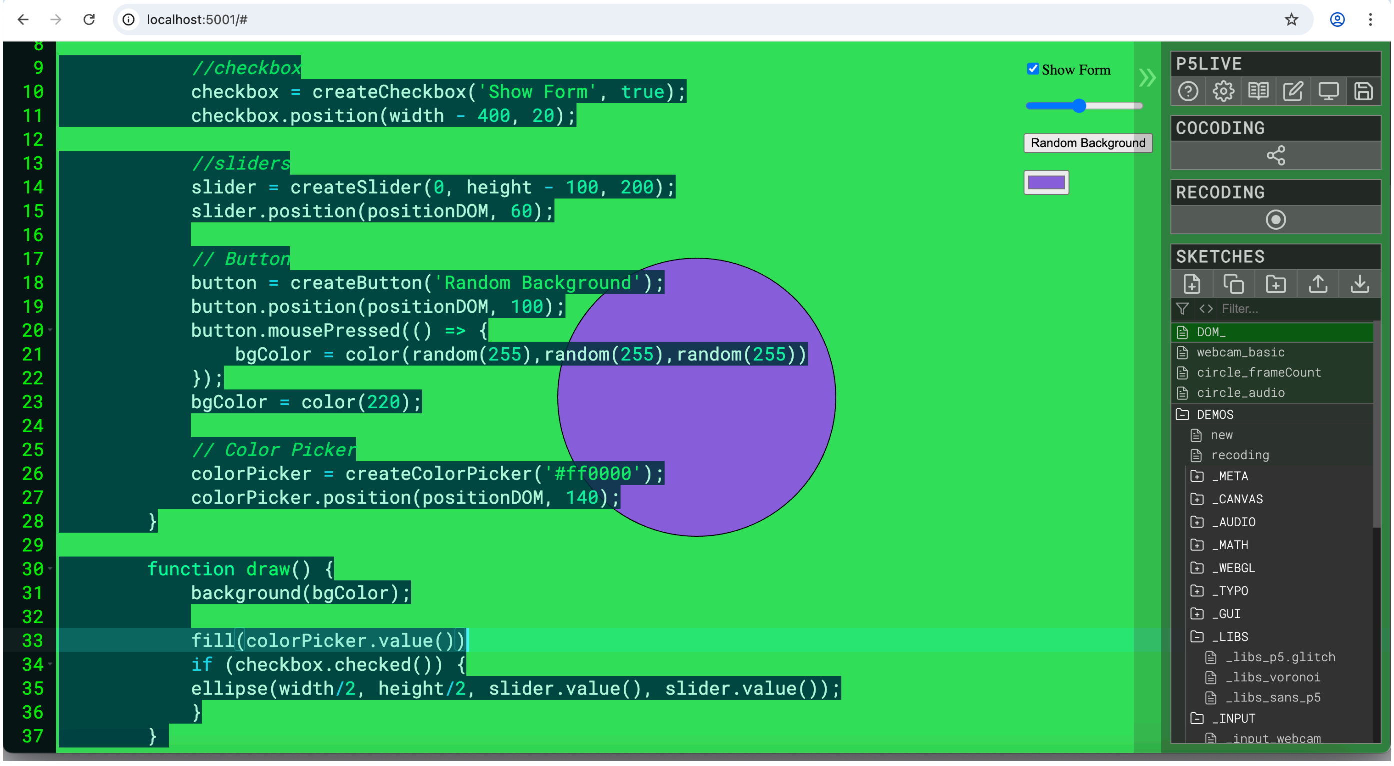


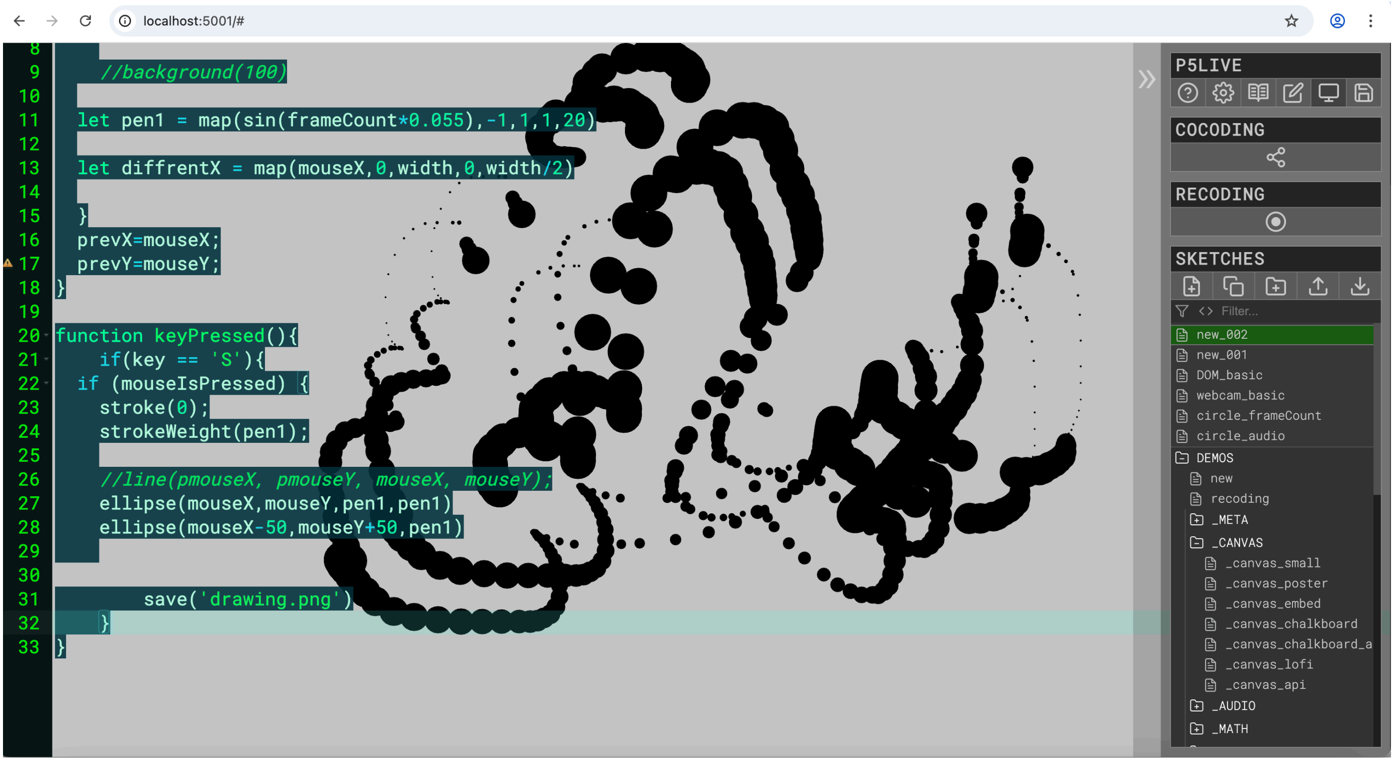
Day 2

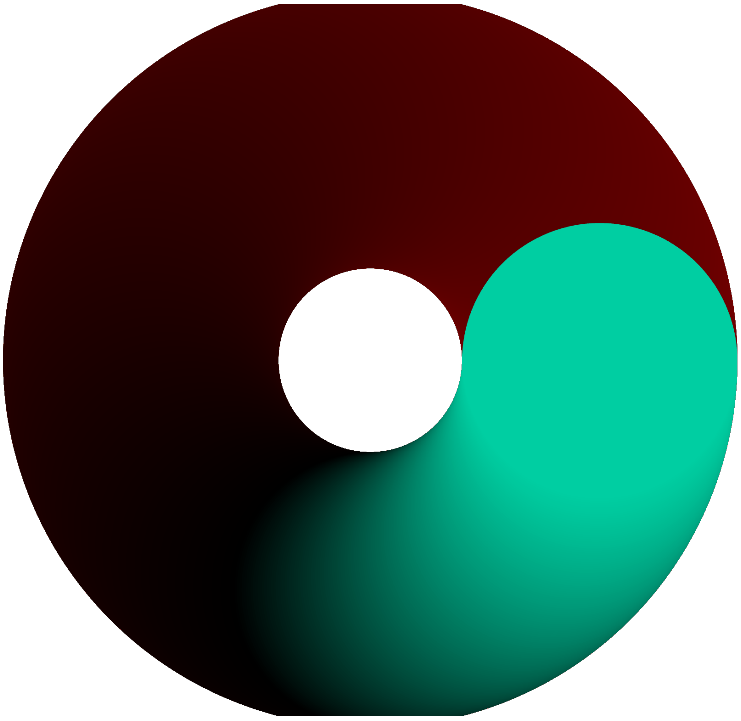
```
TUE, 05/05/26
Alper Yagcioglu, WYSIWYG,
```

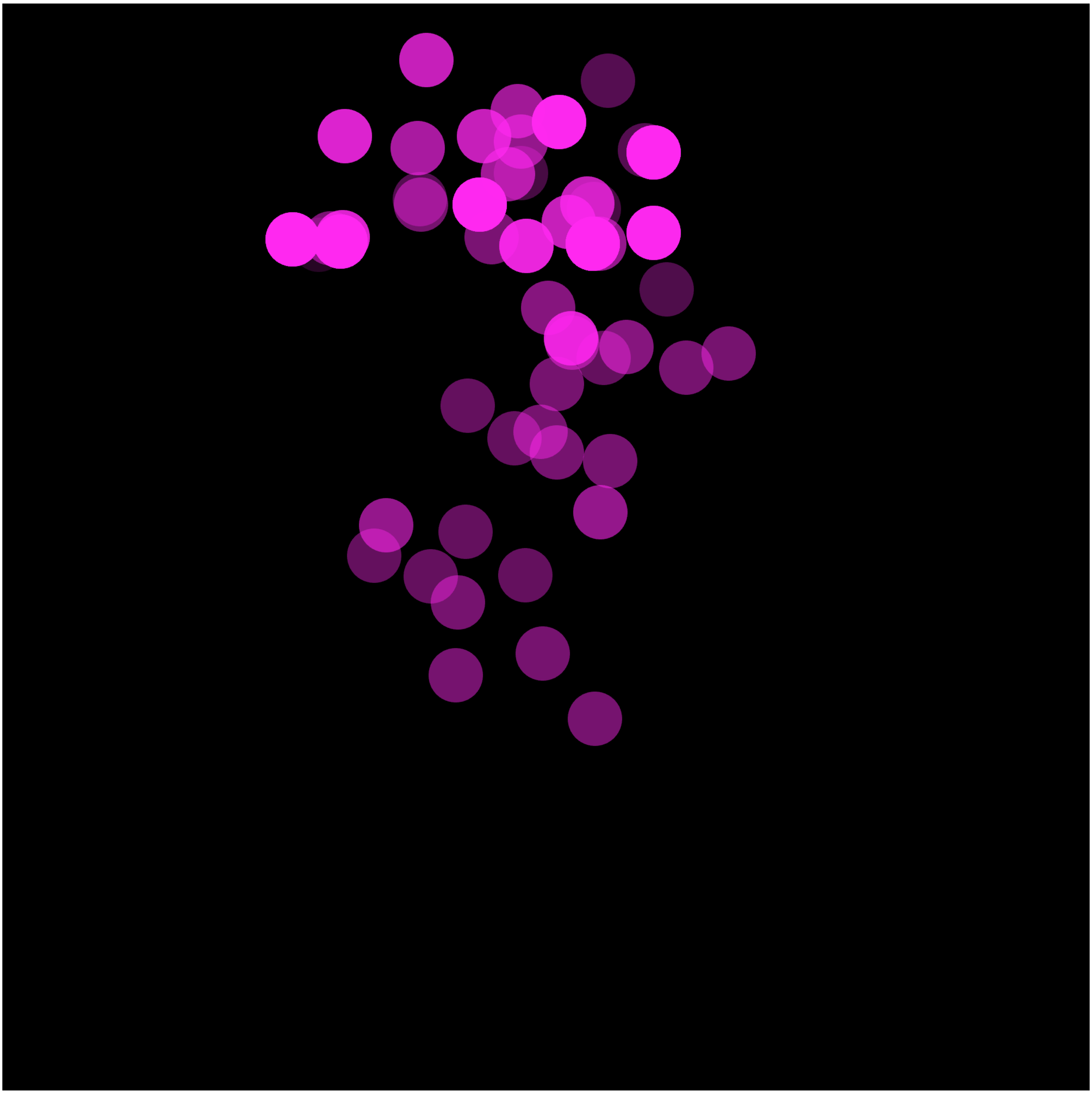
Sutherlands Sketchpad
Muriel Cooper

- GUI (Graphical User Interface)
- HCI (Human-Computer Interaction)
- DOM (Document Object Model)


localhost:5001/#
//checkbox
checkbox = createCheckbox('Show Form', true);
checkbox.position(width - 400, 20);
//sliders
slider = createSlider(0, height - 100, 200);
slider.position(positionDOM, 60);
// Button
button = createButton('Random Background');
button.position(positionDOM, 100);
button.mousePressed(() => {
bgColor = color(random(255),random(255),random(255))
});
bgColor = color(220);
// Color Picker
colorPicker = createColorPicker('#ff0000');
colorPicker.position(positionDOM, 140);
}
function draw() {
background(bgColor);
fill(colorPicker.value())
if (checkbox.checked()) {
ellipse(width/2, height/2, slider.value(), slider.value());
}
}
Show Form
Random Background
P5LIVE
COCODING
RECODING
SKETCHES
Filter...
DOM_
webcam_basic
circle_frameCount
circle_audio
DEMOS
new
recoding
_META
_CANVAS
_AUDIO
_MATH
_WEBGL
_TYPO
_GUI
_LIBS
_libs_p5.glitch
_libs_voronoi
_libs_sans_p5
_INPUT
_input_webcam


localhost:5001/#
//background(100)
let pen1 = map(sin(frameCount*0.055),-1,1,1,20)
let diffrentX = map(mouseX,0,width,0,width/2)
}
prevX=mouseX;
prevY=mouseY;
}
function keyPressed(){
if(key == 'S'){
if (mouseIsPressed) {
stroke(0);
strokeWeight(pen1);
//line(pmouseX, pmouseY, mouseX, mouseY);
ellipse(mouseX,mouseY,pen1,pen1)
ellipse(mouseX-50,mouseY+50,pen1)
save('drawing.png')
}
}
P5LIVE
COCODING
RECODING
SKETCHES
Filter...
new_002
new_001
DOM_basic
webcam_basic
circle_frameCount
circle_audio
DEMOS
new
recoding
_META
_CANVAS
_canvas_small
_canvas_poster
_canvas_embed
_canvas_chalkboard
_canvas_chalkboard_a
_canvas_lofi
_canvas_api
_AUDIO
_MATH





Day 3 + 4

sick...

Day 5

```
FRI, 08/05/26
Paulina Zybinska, Machine Learning
```

Perceptron (Neuron)
Input 1
Input 2
—> Activation function = Result (Output)

What we can do with ml5.js?

BodyPose
Full-body pose estimation

HandPose
Hand-skeleton finger tracking

FaceMesh
Facial landmark detection

ImageClassifier
Image content recognition

SoundClassifier
Audio detection and classification

ml5 NeuralNetwork
Train your own neural networks

ImageClassifer :

```
let libs = ["https://unpkg.com/ml5@1/dist/ml5.min.js", 'https://unpkg.com/hydra-synth',
'includes/libs/hydra-synth.js', 'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js',
'includes/libs/hy5.js']

let fx1 = 0
let fx2 = 0
let fx3 = 0

//sandbox start
```

```
H.pixelDensity(2)
s0.initP5()
P5.toggle(0)

src(s0)
 .modulate(noize(10000), () => fx1)
 .modulateScale(osc(20), () => fx2)
 .add(src(s0).luma(0.9).scale(1.03), () => fx3)
 .luma(() => 0.2 * a.fft[0]) //a.fft[0]
.out()

//sandbox end

//let modelLink = "https://teachablemachine.withgoogle.com/models/KLCzegzC9/"
let modelLink = "https://teachablemachine.withgoogle.com/models/KLCzegzC9/"

// A variable to initialize the Image Classifier
let classifier;

// A variable to hold the video we want to classify
let video;

// Variable for displaying the results on the canvas
let label = "Model loading...";

function preload() {
  classifier = ml5.imageClassifier(modelLink + "model.json");
}

function setup() {
  createCanvas(windowWidth,windowHeight);
  background(255);

  // Using webcam feed as video input, hiding html element to avoid duplicate with canvas
  video = createCapture(VIDEO,{flipped:true});
  video.size(width,height);
  video.hide();
  classifyVideo()
}

function draw() {
  // Each video frame is painted on the canvas
  image(video, 0, 0);
  //background(0)

  // Printing class with the highest probability on the canvas
  fill(255);
  textSize(32);
  text(label, width/2, height/2);

  if(label == "me"){
        circle(width/2, height/2, 100)
      strength = 0.01
      fx1 = 1
      fx2 = 0
      fx3 = 0

    }else if(label == "phone"){
```

```
      rect(width/2, height/2, 200)
      strength = 0.1
      fx1 = 0
      fx2 = 1
      fx3 = 0

    }else {
      triangle(width/2, height/2 + 200, height/2 - 200, width/2 + 400, height/2)
      strength = 0.5
      fx1 = 0
      fx2 = 0
      fx3 = 1
    }

}

// Callback function for when classification has finished
function gotResult(results) {
  // Update label variable which is displayed on the canvas
  label = results[0].label;
  classifyVideo()
}

function classifyVideo() {

  classifier.classify(video,gotResult);
}
```

HandPose :

```
let libs = ['https://unpkg.com/ml5@1/dist/ml5.min.js'];

let video;
let state = 'starting...';
let handPose;
let hands = [];
let options = { maxHands: 2, flipHorizontal: true };
let pScale = 20

let distTip = 0
let distTip2 = 0

function preload() {
  handPose = ml5.handPose(options);
}

function setup() {
  createCanvas(windowWidth, windowHeight);

  video = createCapture(VIDEO, { flipped: true });
  video.size(width, height);
  video.hide();

  handPose.detectStart(video, gotHands);
  state = 'detecting hands';
}

function draw() {
```

```
  background(20, 100);


    for (let i = 0; i < hands.length; i++) {
      const hand = hands[i];
      console.log(hand)
      if(hand === undefined){return}
      for (let j = 0; j < hand.keypoints.length; j++) {
        const kp = hand.keypoints[j];
        const x =  kp.x;
        const y =  kp.y;
        fill(map(j, 0, hand.keypoints.length, 0, 255), 255, map(j, 0, hand.keypoints.length, 255, 0));
        noStroke();
        circle(x, y, pScale * sin(frameCount * 0.1 + j) % pScale);

      }
    }

  noStroke();
  fill(255);
  text(state, 10, height - 10);

   if(hands.length != 0){
    distTip = dist(hands[0].keypoints[4].x, hands[0].keypoints[4].y,
hands[0].keypoints[8].x,hands[0].keypoints[8].y)

    if (hands[1]) {
      distTip2 = dist(hands[1].keypoints[4].x, hands[1].keypoints[4].y,
hands[1].keypoints[8].x,hands[1].keypoints[8].y)
    }
    line(hands[1].keypoints[4].x, hands[1].keypoints[4].y,
hands[1].keypoints[8].x,hands[1].keypoints[8].y)

    stroke(255)
    strokeWeight(5)
    line(hands[0].keypoints[4].x, hands[0].keypoints[4].y,
hands[0].keypoints[8].x,hands[0].keypoints[8].y)

   if(distTip < 50 ){
     circle(hands[0].keypoints[8].x,hands[0].keypoints[8].y,distTip2 * 0.5)
     }
   }
}

function gotHands(results) {
  hands = results;
}
```